OR-AS Operations Research Applications and Solutions	Case Name: **Industrial Complex (2)**	Sector	**Construction (industrial)**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions
Date	**2015**	Project Control	Automatic tracking
File Name	**C2015-07 Industrial Complex (2)**		Tracking based on user input

1. Project description

Project authenticity

An industrial construction project involving site setup, demolition, foundation work, infrastructure, structural construction, utilities, interior/exterior finishes, and landscaping across production halls, offices, storage areas, and specialized facilities.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	138
Planned Duration (PD)	297 days *
Budget At Completion (BAC)	€ 5,999,600
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	27%
Activity Distribution (AD)	38%
Length of Arcs (LA)	0%
Topological Float (TF)	49%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	10.1	8.1	-1.4
CRI-rho	10.0	8.0	-1.4
CRI-tau	7.9	6.5	-1.0

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	24.8	36.7	-1.1
SI	36.6	38.3	-0.6
SSI	3.4	6.9	-0.3
CRI-r	9.3	9.2	-2.0
CRI-rho	9.3	9.2	-1.9
CRI-tau	7.8	6.0	-1.5

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	11.2	8.5
PV - SPI	34.4	32.8
PV - SCI	34.9	32.7
ED - 1	24.5	21.4
ED - SPI	34.4	32.8
ED - SCI	34.5	32.8
ES - 1	6.4	5.7
ES - SPI(t)	25.0	24.8
ES - SCI(t)	25.1	24.9

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1.2	0.4
CPI	2.5	1.1
SPI	21.4	21.4
SPI(t)	17.0	17.0
SCI	21.5	21.5
SCI(t)	17.2	17.2
0.8 CPI + 0.2 SPI	11.5	11.3
0.8 CPI + 0.2 SPI(t)	7.8	7.6

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 9 tracking periods with a length of approximately 2 months week(s). The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	70943.3	-384123.4	-217	1.0	1.0	0.9	0.9
std dev	294208.6	643212.2	266.2	0.1	0.2	0.2	0
final	585056	0	-504	1.1	1.0	0.8	1.0

2.3.3.2. Time forecasting

PD	297 days

Real Duration	360 days

Late	21.2%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	296.6	2.1	2.0	2.0
PV - SPI	297.2	2.1	1.9	1.9
PV - SCI	297.8	2.1	1.9	1.9
ED - 1	298.4	2.1	1.9	1.9
ED - SPI	299.2	1.9	1.9	1.9
ED - SCI	300.0	1.9	1.9	1.9
ES - 1	300.8	1.9	1.8	1.8
ES - SPI(t)	301.6	2.1	1.8	1.8
ES - SCI(t)	302.2	2.1	1.8	1.8

2.3.3.3. Cost forecasting

BAC	€ 5,999,600

Real Cost	€ 5,414,544

Under Budget	-9.8%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	5928656.8	294208.6	1.1	-1.1
CPI	6358100.1	1005453	1.9	-1.9
SPI	5995075.4	659936.3	1.2	-1.2
SPI(t)	5992737.1	714662.2	1.2	-1.2
SCI	6369114.3	703412.4	2.0	-2.0
SCI(t)	6355254.2	670171.5	1.9	-1.9
0.8 CPI + 0.2 SPI	6227436.9	680773.4	1.7	-1.7
0.8 CPI + 0.2 SPI(t)	6218205.4	648211.8	1.6	-1.6